



SARDAR PATEL UNIVERSITY
M. Sc. (IT) INTEGRATED EXAMINATION, V SEM

Date : 23rd Oct. 2021, Saturday
 Session : Evening
 Course No : PS05FIIT01

Time : 02:00pm To 04:00pm
 Sub: Operation Research
 Total marks : 70

Q - 1 Multiple Choice Question

[10]

- i) In graphical representation the bounded region is known as _____ region.
 a) feasible solution b) Solution c) basic solution d) optimal
- ii) Feasible solution satisfies _____.
 a) 'c' & 'b' b) only non-negative restriction
 c) Only constraints d) 'c', 'b' & optimum solution
- iii) In the simplex method the variable leaves the basis if the ratio is.
 a) 0 b) Minimum c) Maximum d) None of them
- iv) The coefficient of artificial variable in the objective function is _____.
 a) -M b) +M c) 0 d) none
- v) For the constraint of greater than equal to type we make use of _____ variable.
 a) Surplus b) Slack c) artificial d) basic
- vi) North - West corner refers to _____.
 a) Top right corner b) Top left corner c) Random minimum d) none
- vii) In non-degenerate solution number of allocated cell is _____.
 a) Equal to $m+n-1$ b) Not equal to $m+n-1$
 c) Equal to $m+n-1$ d) Not equal to $m+n+1$
- viii) _____ operation is carried out on a machine at a time.
 a) Two b) Only One c) At least one d) None of them
- ix) Merge event represents _____ of two or more events.
 a) beginning b) Completion c) Splitting d) None of them
- x) Activity which starts only after finishing other activity is called _____.
 a) dummy b) Successor c) Predecessor d) None of them

Q - 2 Fill in the Blanks and True - False

[08]

- i) Operation research is the application of _____ methods to arrive at the optimal solutions to the problems.
- ii) The linear function of variables which is to be maximized or minimized is called objective function. (True / False)
- iii) The _____ variable is added to the constraint of less than equal to type.
- iv) The coefficient of slack variable in the objective function is 0. (True / False)
- v) The _____ method's solution for transportation problem is sometimes an optimal solution itself.
- vi) The Hungarian method used to obtain optimum solution of travelling salesman. (True / False)
- vii) _____ operation is carried out on a machine at a time.
- viii) Burst event represents beginning of more than one activities. (True / False)

①

P.T.O.

Q – 3 Short Answer **attempt any ten (Each carry 2 marks)**

[20]

- i) Define Operation Research and feasible solution.
- ii) Define Unbounded and Optimum Solution.
- iii) State phases for formulating the operations research problems.
- iv) List the advantages of LPP.
- v) How will you construct a mathematical model.
- vi) Define slack variables.
- vii) What is transportation problem
- viii) What is feasible solution, Basic feasible solution and optimum solution?
- ix) What is assignment problem?
- x) How will you allocate jobs in a sequence if two jobs on first machine have same processing time?
- xi) What is Total float?
- xii) What is successor activity?

Q – 4 Attempt any four **(Each carry 8 marks)**

[32]

- i) Write different applications of Operations Research.
- ii) Explain different types of model for Operations Research.
- iii) Write steps of the algorithm for solving LPP by the simplex method.
- iv) Solve following by simplex method.

$$\text{Maximize } Z = 5x_1 + 3x_2$$

$$\text{s.t. } 3x_1 + 5x_2 \leq 15,$$

$$5x_1 + 2x_2 \leq 10,$$

$$x_1, x_2 \geq 0.$$
- v) Write algorithm and steps to solve transportation problem by NWCM and LCM.
- vi) Find the initial basic feasible solution of the following transportation problem using VAM method and check its optimality too using appropriate method

Warehouse / Factory	W1	W2	W3	W4	Factory Capacity
F1	19	30	50	10	7
F2	70	30	40	60	9
F3	40	8	70	20	18
Requirement	5	8	7	14	

- vii) Write general assumption and procedure for solving problem of sequencing with two machines.
- viii) Find the critical path and calculate the Total float and Free float for the following PERT diagram.

